



Panchip Microelectronics Co., Ltd.

PAN3029 Crystal Selection Guide

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Document Description

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Revision History

Version	Revision Date	Description
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1 Introduction

1.1 Background Overview

PAN3029 is a wireless transceiver chip used for long-distance communication. The chip has a minimum sensitivity of -143dBm, a maximum transmit power of 20dBm, and an industry-leading link budget of 163dB. At a transmit power of 20dBm, the chip generates a lot of heat, which is transferred to the crystal through the PCB. Since the frequency of the crystal changes with temperature, frequency drift will occur during the transmission process. This drift will pose a great challenge to the receiving demodulation. In severe cases, it will lead to receiving demodulation data errors and packet loss.

2 Frequency drift mechanism and influencing factors

Crystal frequency drift is mainly related to the temperature characteristics of the crystal itself and the temperature change. The crystal temperature change depends on the heat dissipation of the chip and the heat change from the chip to the crystal through the PCB. The heat dissipation of the chip is mainly related to the transmission power, while the heat change from the chip to the crystal is mainly related to the PCB size and the heat insulation between the crystal and the chip.

The reception tolerance to frequency drift is related to BW, SF, LDR, Payload length and working frequency band. For different configurations, each symbol has different requirements for frequency drift, and the total transmission time of a frame is also different, resulting in different requirements for frequency drift.

The following is a detailed description of each influencing factor.

2.1 Power dissipation

The following is the power dissipation at 20dBm, 14dBm, and 10dBm in the 490MHz frequency band. The calculation formula for power dissipation is: power supply voltage VCC, chip current is I, input power is Pin, output power is Pout, and power dissipation is Pd.

$$Pd = Pin - Pout = VCC * I - Pout$$

VCC (V)	I (mA)	Pi (mW)	Pout (dBm)	Pout (mW)	Pd (mW)
3.3	91	300.3	20	100	200.3
3.3	50	165	14	25.1	139.8
3.3	30	99	10	10	89

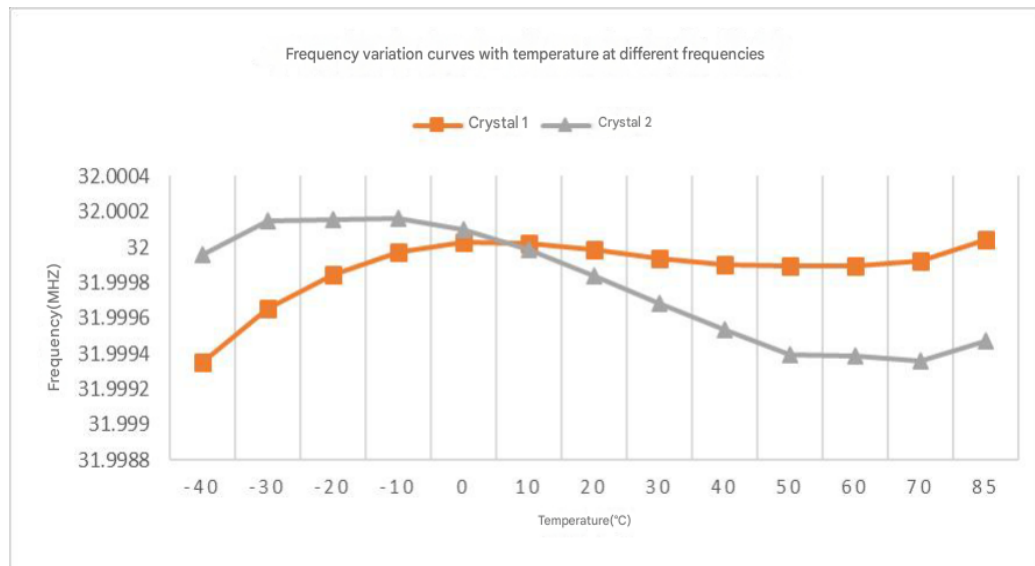
From the table above, we can see that 10dBm will reduce the power dissipation by more than half compared to 20dBm.

2.2 Crystal frequency characteristics

The following tests are conducted on 32M crystals from different manufacturers. By observing the change of single carrier frequency over time in the 490MHz band and counting the drift within a certain period of time after the single carrier is launched, it can be seen that the drift of different crystals varies greatly. The ones with smaller frequency drift are 1C232000AA0B from Big Vacuum and FSX-3M from Fuji.

厂家	料号	立创编号	封装形式	负载电容 (pF)	温度特性 (-20~70°) ppm)	频率精度 (ppm)	频率漂移 (Hz)
大真空	1C232000AA0B	/	SMD3225-4P	12	10	10	250
泰晶	SX32Y032000B91T	C271639	SMD3225-4P	9	10	10	342
泰晶	SX32Y032000BC1T	C271658	SMD3225-4P	12	10	10	496
安基	CXA-032000-3F7E40	C369204	SMD3225-4P	12	15	10	800
晶技	7M32000020	C5203632	SMD3225-4P	12	/	/	738
爱普生	X1E000021016900	C255932	SMD3225-4P	12	10	10	488
晶光华	S3232000091040	C426986	SMD3225-4P	12	30	30	460
晶科鑫	7U32000E12UCG	C251626	SMD3225-4P	12	10	10	528
川晶	C3SB32.000F12E12	C1122385	SMD3225-4P	12	20	10	500
长晶	CJ13-320001210A20	C712874	SMD3225-4P	12	20	10	490
晶优	JYXT32S4-032.00000-93C4B0	C357891	SMD3225-4P	12	20	10	760
浩都	7B032000M01	C655025	SMD3225-4P	12	20	10	384
台湾加高	X3S032000B91H-HU	C254375	SMD3225-4P	9	20	10	716
晶威特	CM4032M00012T8188141	C2932987	SMD3225-4P	12	15	10	330
晶威特	CF4032M00012T5074007	C16197273	SMD3225-4P	12	20	10	292
晶威特	CF4032M00012T5074007	C5765975	SMD3225-4P	12	20	10	850
晶宝	01X.MD.110HJV0032000000	C7570525	SMD3225-4P	10	10	10	380
富士	FSX-3M	/	SMD3225-4P	12	/	10	224
晶威特	CN4032M00009T8188116	C2932962	SMD 2016-4P	9	30	10	522
晶威特	CN4032M00012T8188117	C2932963	SMD 2016-4P	12	15	10	516
鸿星	E3SB32E00000BE	C5241172	SMD3225-4P	10	10	10	470
索斯特	2520+32M+9PF+10PPM	C5181488	SMD3225-4P	12	/	10	670

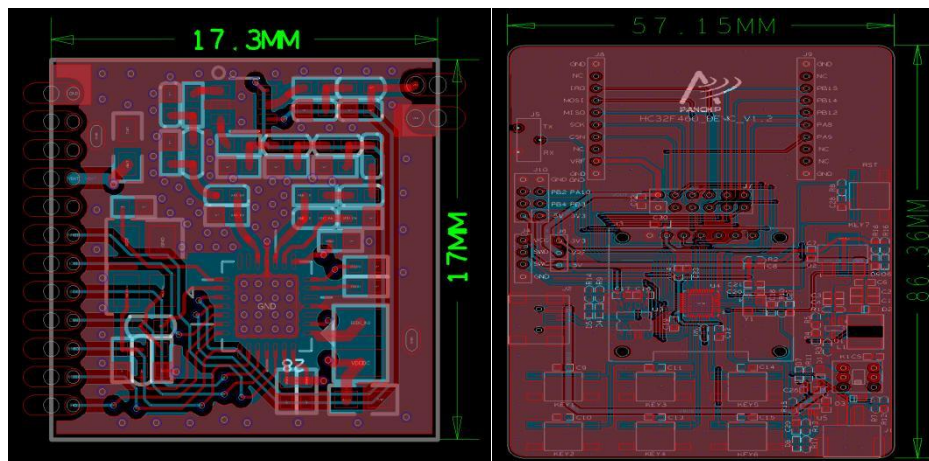
The figure below shows the temperature characteristics of two crystals. The frequency drift of crystal 1 is 250Hz, and the frequency drift of crystal 2 is 850Hz. It can be seen from the curve that the frequency characteristic stability of crystal 1 is better in the range of -10~70°C, while that of crystal 2 is worse. The test results of temperature characteristics and frequency drift are consistent.



2.3 PCB design drift impact

Comparing the frequency drift of the same crystal on two PCBs with different sizes, it is found that the larger the PCB, the better the heat dissipation and the smaller the frequency drift. Under the same time, the small module drifts to 850Hz, while the R&D EVB test board drifts to only 288Hz.

This shows that the larger the board, the better the heat dissipation and the smaller the impact on the crystal temperature. In addition to size, it is also important to do a good job of PCB thermal isolation of the crystal and chip, which needs to be considered in PCB thermal design.



2.4 Transmission time

For different BW and SF, the duration of each symbol is different, and the total packet transmission time will also be different, which can be calculated using the PAN3029 chip calculator.

PAN3029 Chip Calculator			
Input parameters	value	unit	describe
Payload bytes (2 to 255)	64	Byte	User payload length
bandwidth	125.0	KHz	-
SF factor	5	-	-
Carrier frequency (150MHz to 1.2 GHz)	470	MHz	-
noise figure	6	dB	The default noise figure is 6
Transmit output power (-7 to +22)	22.0	dBm	Radiated emission power
Message mode	1	-	Support messages by default
Code rate	1	-	1=4/5, 2=4/6, 3=4/7, 4=4/8
Number of preamble symbols	8	-	The default value is 8, and the input range is 8 to 65535
low rate mode	0	-	1 = Supports low-rate mode, 0 = Does not support low-rate mode
Output parameters			
	value	unit	describe
Receive sensitivity	-119.5	dBm	-
Link Budget	142	dB	-
Symbol length	0.3	ms	-
Preamble duration	3.6	ms	-
Payload duration	36.6	ms	-
Total delivery time	40.3	ms	Contains CRC16
Actual data rate	12.72	kbps	Includes preamble/header/CRC

The following lists and calculates the symbol time and total time for sending a frame for different SF and BW. The smaller the BW, the larger the SF, the longer the symbol time, the longer the total time for a frame, the greater the frequency drift, and the higher the requirement for frequency stability.

BW	SF	CR	Preamble数量 (bit)	Payload数量 (Bytes)	符号时间 (ms)	一帧总时间 (ms)
125	5	1	20	64	0.3	40.3
125	6	1	20	64	0.5	67.7
125	7	1	20	64	1	118
125	8	1	20	64	2	215.6
125	9	1	20	64	4.1	390.1
125	10	1	20	64	8.2	698.4
125	11	1	20	64	16.4	1314.8
500	12	1	20	64	8.2	616.4
250	12	1	20	64	16.4	1232.9
125	12	1	20	64	32.8	2465.8
62.5	12	1	20	64	62.5	4931.6

2.5 Carrier frequency impact

The carrier frequency f_{TX} of TX end is $f_{REFTX} \cdot (n_{TX} + m_{TX})$, the carrier frequency f_{RX} of RX end is $f_{REFRX} \cdot (n_{RX} + m_{RX})$, the absolute value of the intermediate frequency f_{IF} of RX end is $f_{TX} - f_{RX}$. The frequency drift during the transmission process of TX end affects the frequency drift of the intermediate frequency of RX end. The magnitude of the frequency drift is related to the stability of the crystal f_{REFTX} of TX end and the multiple $n_{TX} + m_{TX}$. And $n_{TX} + m_{TX}$ is related to the operating frequency, that is, the higher the frequency, the greater the impact of f_{REFTX} drift. For the same crystal, the same output power, assuming the power dissipation is the same, the frequency drift of 866M is twice that of 433M.

3 Crystal Recommendation

3.1 Passive Crystal

Based on Panqi's DTM-CC0015 module, the operating frequency band is 408~510MHz, the operating temperature is -20~70°C, the transmission payload is no more than 20Bytes, the transmission power is 20dBm, and the BW, SF, and LDR configurations are shown in the following table. Passive crystals with good temperature characteristics can be used, and the frequency accuracy is required to be 10ppm, and the temperature stability of -20~70°C is 10ppm. KDS's 1C232000AA0B

The test effect will be better than other passive crystals. If the cost is acceptable, it is recommended to use KDS's 1C232000AA0B.

BW (KHz)	SF	LDR
500	5	OFF
500	6	OFF
500	7	OFF
500	8	OFF
500	9	OFF
500	10	OFF
500	11	OFF
500	12	OFF
250	5	OFF
250	6	OFF
250	7	OFF
250	8	OFF
250	9	OFF
250	10	OFF
250	11	OFF
125	5	OFF
125	6	OFF
125	7	OFF
125	8	OFF
125	9	OFF
125	10	OFF
62.5	5	OFF
62.5	6	OFF
62.5	7	OFF
62.5	8	OFF
62.5	9	OFF

Based on the above conditions, changing the power, temperature, frequency band and payload length will change the BW and SF configuration of the available passive crystal. It may increase the available range by one gear at the lowest rate (BW decreases by one gear, or SF increases by one gear), or it may decrease the available range by one gear at the lowest rate (BW is too high by one gear, or SF decreases by one gear). If multiple conditions change, the effects of multiple gear changes will be superimposed. For example:

Note 1: When the power is reduced to about 10dBm, 125K SF11 can use passive crystals;

Note 2: When the operating temperature reaches -40°C, 125K SF10 and 62.5 SF9 cannot use passive crystals;

Note 3: When the operating frequency band is 863~870MHz and 902~928MHz, 125K SF10 and 62.5 SF9 cannot use passive crystals;

Note 4: When the Payload length is greater than 20Bytes and less than 70Bytes, 125K SF10 and 62.5 SF9 cannot use passive crystals.

3.2 Temperature Compensated Crystal Oscillator

Configurations that cannot use passive crystals require the use of 1ppm active temperature-compensated crystals, and all operating temperatures must meet the 1.5ppm requirement. For applications between -40 and 85°C, KDS's active temperature-compensated crystal 1XXD32000MBA is recommended.

($T_A = -40 \sim +85^\circ\text{C}$, $L_{\text{OAD_R}}/C = 10\text{k}\Omega/10\text{pF}$, $V_{\text{CC}} = +3.3\text{V}$, unless otherwise noted)

	Item	Conditions	Limits			unit	Notes
			min.	typ.	max.		
1	Current Consumption		-	-	+2.0	mA	
2	Output Level		0.8	-	-	V _{P-P}	1
3	Symmetry	GND level (DC cut)	40/60	-	60/40	%	
4	Harmonics		-	-	-5	dBc	
5	Frequency Stability						
	1.Tolerance	After 2 times reflow Ref. to nominal frequency	-	-	±1.5	ppm	2,3
	2.vs Temperature	$T_A = -30 \sim +85^\circ\text{C}$ Ref. to frequency ($T_A = +25^\circ\text{C}$)	-	-	±0.5	ppm	
		$T_A = -40 \sim -30^\circ\text{C}$ Ref. to frequency ($T_A = +25^\circ\text{C}$)	-	-	±1.0	ppm	
	3.vs Supply Voltage	$V_{\text{CC}} = +3.3\text{V} \pm 5\%$	-	-	±0.2	ppm	
	4.vs Load Variation	$L_{\text{OAD_R}}/C = (10\text{k}\Omega/10\text{pF}) \pm 10\%$	-	-	±0.2	ppm	
	5.vs Aging	$T_A = \text{Room ambient}$	-	-	±1.0	ppm/year	
6	Start Up Time	@90% of final V _{out} level	-	-	2.0	ms	
7	SSB Phase Noise	Relative to F0 level offset 1kHz	-	-	-125	dBc/Hz	